

World Trade Center-Cardiorespiratory and Vascular Dysfunction:
Assessing the Phenotype and Metabolome of a Murine Particulate Matter Exposure
Model

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Table 1. Echocardiography: 24-Hours and 1-Month After WTC-PM Exposure in Comparison to Their Controls

Parameters		24-Hours		1-Month	
		WTC-PM	PBS	WTC-PM	PBS
PA	PAT (ms)	10.83 (3.15)**	17.57 (3.25)	11.64 (3.01)**	17.34 (3.02)
	PET (ms)	56.37 (5.82)	51.11 (6.91)	56.55 (10.95)	56.21 (6.06)
	PAT/PET	0.19 (0.06)***	0.34 (0.03)	0.21 (0.03)***	0.29 (0.02)
	PA VTI (mm)	14.04 (3.11)	16.12 (2.88)	17.49 (2.71)	19.45 (4.45)
	Peak Pressure (mmHg)	0.87 (0.28)*	1.22 (0.33)	1.06 (0.22)*	1.32 (0.20)
	MAP (Common) (mmHg)	74.22 (1.41)**	70.81 (1.65)	73.76 (1.35)**	71.46 (0.95)
	MAP (PAT<120ms) (mmHg)	83.41 (1.95)**	79.02 (2.02)	82.79 (1.87)**	79.62 (1.31)
	Peak Velocity (mm/s)	471.10 (65.98)	537.75 (71.5)	511.50 (57.53)*	573.60 (44.9)
	Mean Velocity (mm/s)	267.13 (39.95)*	314.70 (39.06)	312.30 (28.73)	336.50 (30.4)
	Peak Gradient (mmHg)	0.86 (0.24)*	1.15 (0.29)	1.08 (0.18)	1.29 (0.24)
	Mean Gradient (mmHg)	0.29 (0.08)*	0.40 (0.09)	0.39 (0.06)	0.46 (0.08)
LV	HR (beats/min)	474.00 (23.13)	478.37 (38.8)	450.78 (31.24)	476.05 (13.9)
	CO (ml/min)	10.74 (2.75)*	13.99 (4.26)	13.78 (1.72)**	17.16 (1.48)
	SV (μl)	22.58 (5.45)*	29.01 (8.01)	30.67 (4.02)	31.57 (8.63)
	EF (%)	74.06 (11.59)	79.71 (9.63)	66.01 (6.18)	71.26 (9.47)
	FS (%)	42.80 (8.75)	48.06 (9.1)	35.70 (4.69)	40.47 (9.41)
	Volume _{Systolic} (μl)	7.50 (2.83)	8.16 (7.2)	16.30 (4.93)	13.80 (5.93)
	Volume _{Diastolic} (μl)	30.08 (6.23)	37.17 (13.53)	46.97 (7.78)	45.37 (14.0)
	Diameter _{Systolic} (mm)	1.59 (0.24)	1.58 (0.45)	2.18 (0.27)	1.99 (0.49)
	Diameter _{Diastolic} (mm)	2.80 (0.24)	3.03 (0.42)	3.37 (0.24)	3.29 (0.48)
Aorta	Aortic Root Diameter (mm)	1.41 (0.05)	1.42 (0.08)	1.60 (0.04)	1.58 (0.08)
	Diameter _{Systolic} (mm)	1.34 (0.08)	1.33 (0.11)	1.15 (0.18)	1.11 (0.23)
	Diameter _{Diastolic} (mm)	1.63 (0.00)	1.67 (0.04)	1.61 (0.12)	1.42 (0.15)
	AoV AAT (ms)	13.47 (2.81)	19.03 (5.81)	13.29 (1.59)****	17.75 (4.37)
	AoV AET (ms)	55.69 (3.47)*	50.69 (1.61)	59.35 (6.11)*	50.97 (3.82)
	AoV AAT/AET (ratio)	0.24 (0.04)*	0.37 (0.11)	0.22 (0.01)****	0.35 (0.09)
	AoV VTI (mm)	22.27 (4.88)*	15.82 (4.63)	24.30 (4.31)*	18.04 (3.68)
	AoV Peak Velocity (mm/s)	679.70 (119.6)	528.93 (127.6)	688.30 (73.01)*	824.54 (109.92)
TV	A Peak (mm/s)	225.41 (2.35)	216.43 (16.05)	323.80 (34.48)**	248.45 (21.49)
	E Peak (mm/s)	150.85 (3.95)	152.75 (3.35)	221.03 (9.08)****	151.25 (10.32)
	E/A (ratio)	0.67 (0.01)	0.71 (0.04)	0.72 (0.03)*	0.61 (0.02)
RV	Internal Diameter (mm)	1.02 (0.09)	0.97 (0.15)	0.84 (0.04)**	0.98 (0.04)

Mean ± SD, n=8/group,

*P<0.05 **P<0.01 ***P<0.001 ****P<0.0001 and others are not significant.

Abbreviations: PA Pulmonary Artery; LV Left Ventricle; RV Right Ventricle; PAT Pulmonary Acceleration Time; PET Pulmonary Ejection Time; VTI Velocity Time Integral; HR Heart Rate; CO Cardiac Output; EF Ejection Fraction; FS Fractional Shortening; AoV Aortic Valve; TV Tricuspid Valve; E Peak The Peak Velocity of E (early) Wave; A Peak The Peak Velocity of A (atrial/after) Wave

Supplemental Table Legend

Table 1. Echocardiographic results, 24-hrs and 1-M after WTC-PM and PBS Exposure. All values Mean $\pm$ SD, n=8, *p<0.05; **p<0.01; ***p<0.001; ****p<0.0001 and others are not significant.